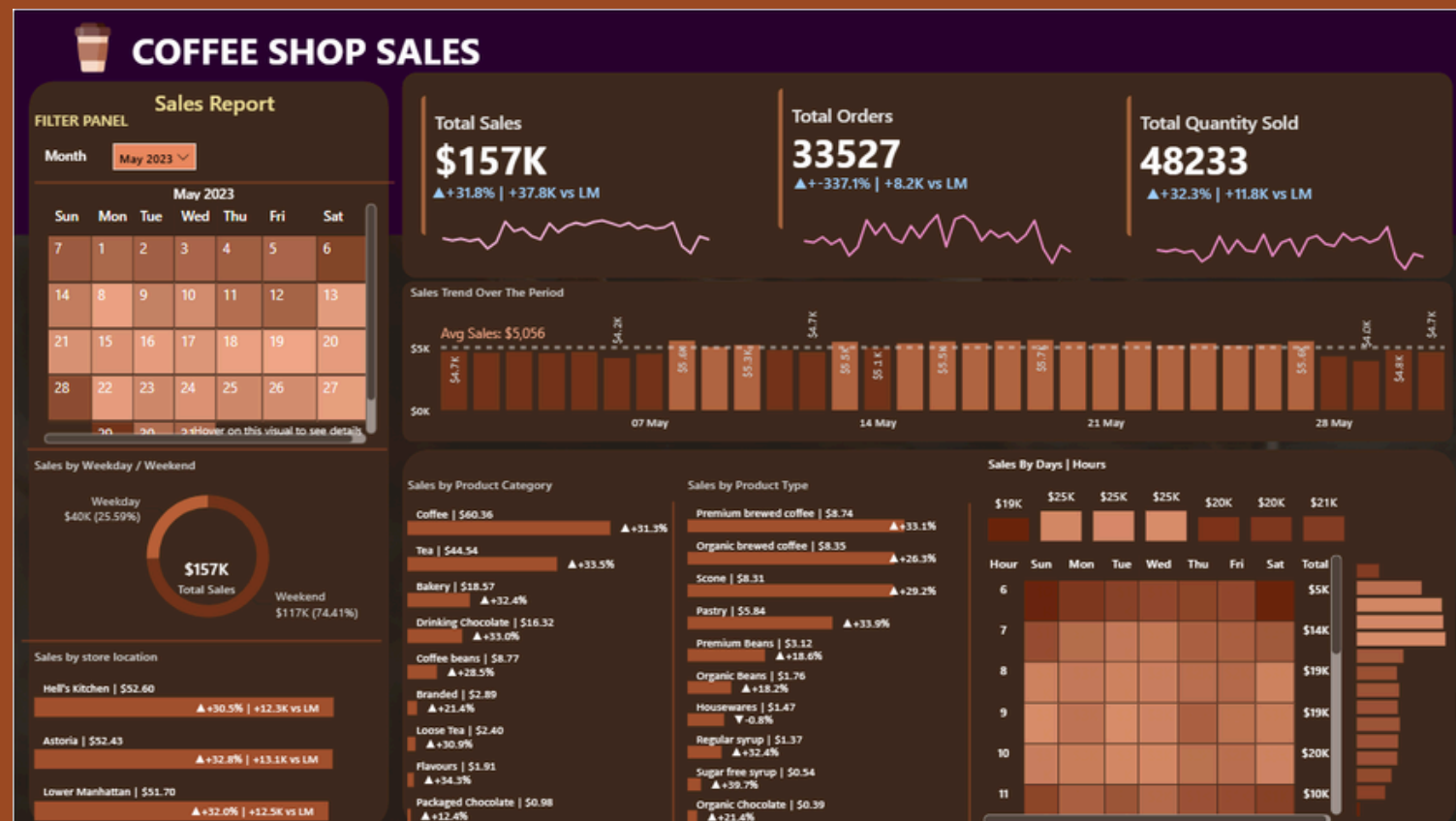


DATA ANALYST PORTFOLIO PROJECT



SQL + POWER BI START TO END



```
48 SELECT
49     MONTH(transaction_date) AS month,
50     ROUND(SUM(unit_price * transaction_qty)) AS total_sales,
51     (SUM(unit_price * transaction_qty) - LAG(SUM(unit_price * transaction_qty), 1)
52     OVER (ORDER BY MONTH(transaction_date))) / LAG(SUM(unit_price * transaction_qty), 1)
53     OVER (ORDER BY MONTH(transaction_date)) * 100 AS mom_increase_percentage
54 FROM
55     coffee_shop_sales
56 WHERE
57     MONTH(transaction_date) IN (4, 5) -- for months of April and May
58 GROUP BY
59     MONTH(transaction_date)
60 ORDER BY
61     MONTH(transaction_date);
62
63 SELECT COUNT(*) AS Total_Rows
64 FROM coffee_shop_sales;
65
66
67 SELECT COUNT(transaction_id) AS Total Sales
```

PART - 1

MYSQL



FIRING SQL QUERIES TO SOLVE THE BUSINESS PROBLEMS



```
77 SELECT
78     MONTH(transaction_date) AS month,
79     ROUND(COUNT(transaction_id)) AS total_orders,
80     (COUNT(transaction_id) - LAG(COUNT(transaction_id), 1)
81     OVER (ORDER BY MONTH(transaction_date))) / LAG(COUNT(transaction_id), 1)
82     OVER (ORDER BY MONTH(transaction_date)) * 100 AS mom_increase_percentage
83 FROM
84     coffee_shop_sales
85 WHERE
86     MONTH(transaction_date) IN (4, 5) -- for April and May
87 GROUP BY
88     MONTH(transaction_date)
89 ORDER BY
90     MONTH(transaction_date);
91
92 • SELECT SUM(transaction_qty) AS Total_quantity_sold
93 FROM coffee_shop_sales
94 WHERE
95     MONTH(transaction_date) = 5 -- May Month;
96
```

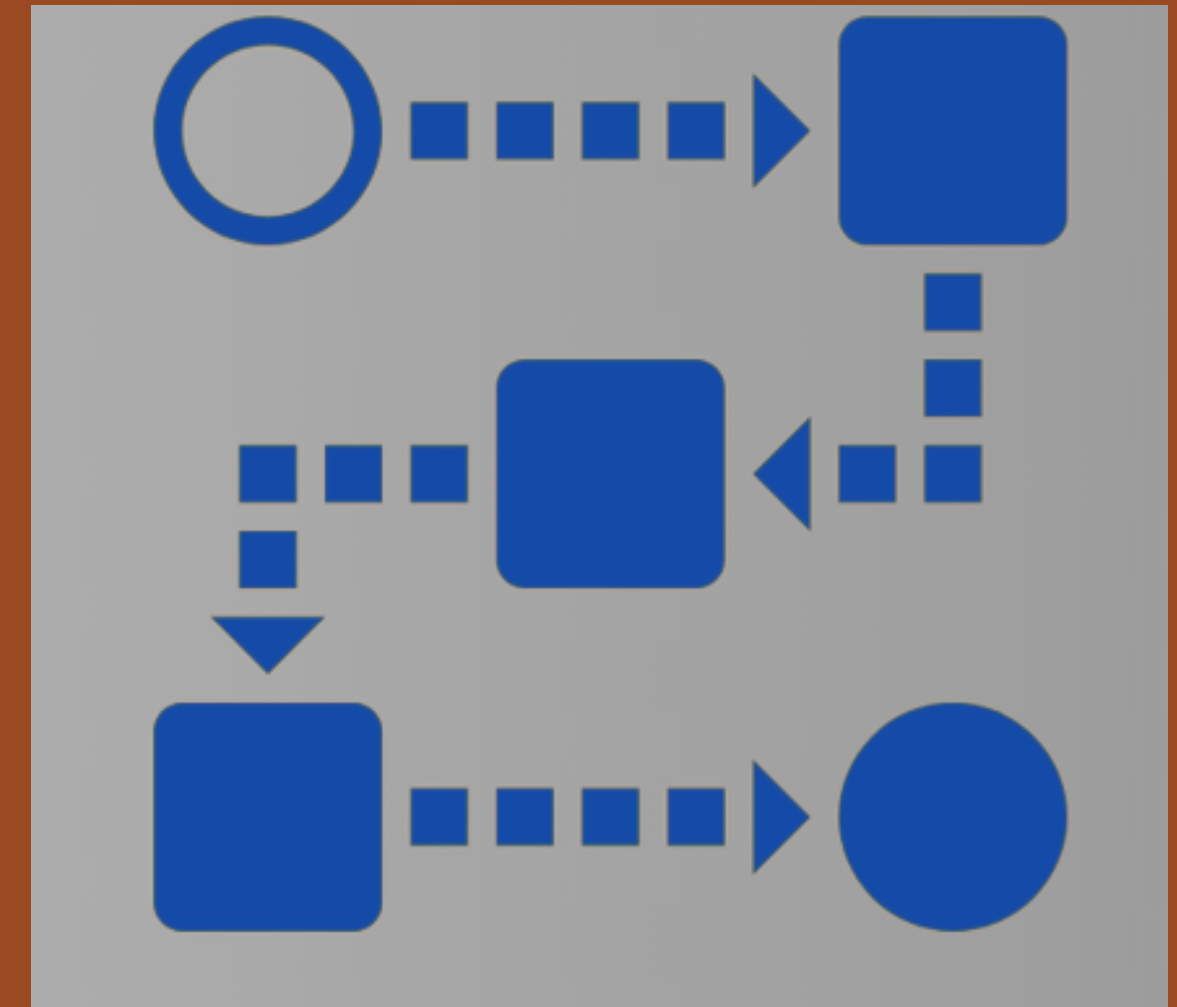
Output :

#	Time	Action	Message
1	00:03:58	SELECT COUNT(*) AS Total_Rows FROM coffee_shop_sales LIMIT 0, 1000	1 row(s) returned



STEPS FOR MY SQL

- Data Walkthrough
- Raw Data File Preparation
- Creating Database
- Importing File
- Cleaning Imported File
- Changing Data Types
- FIRING SQL QUERIES TO SOLVE THE BUSINESS PROBLEMS
- Storing Result
- Preparaing SQL Documents



PART - 3

MY SQL



FUNTIONALITIES

STR_TO_DATE

ROUND

SUM

COUNT

AVG

LAG

MONTH

DAY

DAYOFWEEK

SELECT

ALIAS

MAX/MIN

HOUR

ALTER TABLE

UPDATE TABLE

CHANGE COLUMN

WHERE

GROUP BY

ORDER BY

CASE

LIMIT

WINDOW

FUNCTIONS

JOINS

SUBQUERIES

PROBLEM STATEMENT



KPI'S REQUIREMENTS

1. Total Sales Analysis:

- Calculate the total sales for each respective month.
- Determine the month-on-month increase or decrease in sales.
- Calculate the difference in sales between the selected month and the previous month.

2. Total Order Analysis:

- Calculate the total number of orders for each respective month.
- Determine the month-on-month increase or decrease in the number of orders.
- Calculate the difference in the number of orders between the selected month and the previous month.

3. Total Quantity sold Analysis:

- Calculate the total quantity sold for each respective month.
- Determine the month-on-month increase or decrease in the total quantity sold.
- Calculate the difference in the total quantity sold between the selected month and the previous month.

PROBLEM STATEMENT



CHARTS REQUIREMENTS

1. Calender Heat Map:

- Implement a calendar heat map that dynamically adjusts based on the selected month from a slicer.
- Each day on the calendar will be color-coded to represent sales volume, with darker shades indicating higher sales.
- Implement tooltip to display detailed metrics (Sales , Orders , Quantity) when hovering over a specific day.

2. Sales Analysis By Weekdays And Weekends:

- Segment sales data into weekdays and weekends to analyze performance variations.
- Provide insights into whether sales patterns differ significantly between weekdays and weekends.

3. Sales Analysis By Store Locaton:

- Visualize sales data by diffrent store locations.
- Include month-over-month (MOM) difference metrics based on the selected month in the slicer.
- highlight MOM sales increase or decrease for each store location to identify trends.

PROBLEM STATEMENT



CHARTS REQUIREMENTS

4. Daily Sales Analysis With Average Line:

- Display daily sales for the selected month with a line chart.
- Incorporate an average line on the chart to represent the average daily sales.
- Highlight bars exceeding or falling below the average sales to identify exceptional sales days.

5. Sales Analysis By Product Category:

- Analyze sales performance across different product categories.
- Provide insights into which product categories contribute the most to overall sales.

6. Top 10 Products By Sales:

- Identify and display the top 10 products based on sales volume.
- Allow users to quickly visualize the best-performing products in terms of sales.

7. Sales Analysis By Days And Hours:

- Utilize heat map to visualize sales patterns by days and hours.
- Implement tooltips to display detailed (Sales , Orders , Quantity) when hovering over a specific day-hour.